



RegPulse

AI-Powered Environmental Intelligence
AI
for the European Shipping Industry

Developers **Andreas,**
Gretel, Vincent

Timeframe **End of Business Day**
(Rapid Prototype)

Core Stack **Gradio + OpenAI GPT-4o-mini**
+ OpenAI TTS

MEET THE TEAM



Andreas



News Retrieval Lead

Project idea. News Retrieval implementation, NewsAPI integration and initial query design.



Gretel



AI & Integration Lead

Overall application architecture, module integration, Prompt Engineering, Search Intelligence strategy, Gradio interface, UX improvements, testing, debugging and final product integration.



Vincent



Audio Architecture Lead

Text-to-Speech implementation and OpenAI audio generation.

Building RegPulse AI required structured, collaborative parallel processing. Our team distributed responsibilities according to our modular system pipeline.

Integrated Operations: Modules designed with clean boundaries for fast, independent development.

Git Mastery: Managed repository branches to seamlessly combine scraping, AI prompt pipelines, and audio conversion.

THE IDEA

Compliance and Relevance Pressure

The European maritime industry operates in a rapidly changing regulatory environment. Environmental policies, emissions regulations, and sustainability initiatives are updated frequently.

Managers and decision-makers often lack the time to read lengthy reports and news articles. RegPulse AI addresses this challenge by automatically collecting relevant information, summarizing it, explaining its business implications, and converting it into an easy-to-listen podcast.

POLICY EVOLUTION

24/
7

Overwhelmed Executives: Busy maritime decision-makers lack the time to review hundreds of pages of daily regulatory changes.

Compliance Penalties: Failure to immediately operationalize updated directives leads to severe environmental surcharges.

Overwhelming Technicality: Traditional legal-heavy publications lack quick business insights.

INTELLIGENT SOLUTION

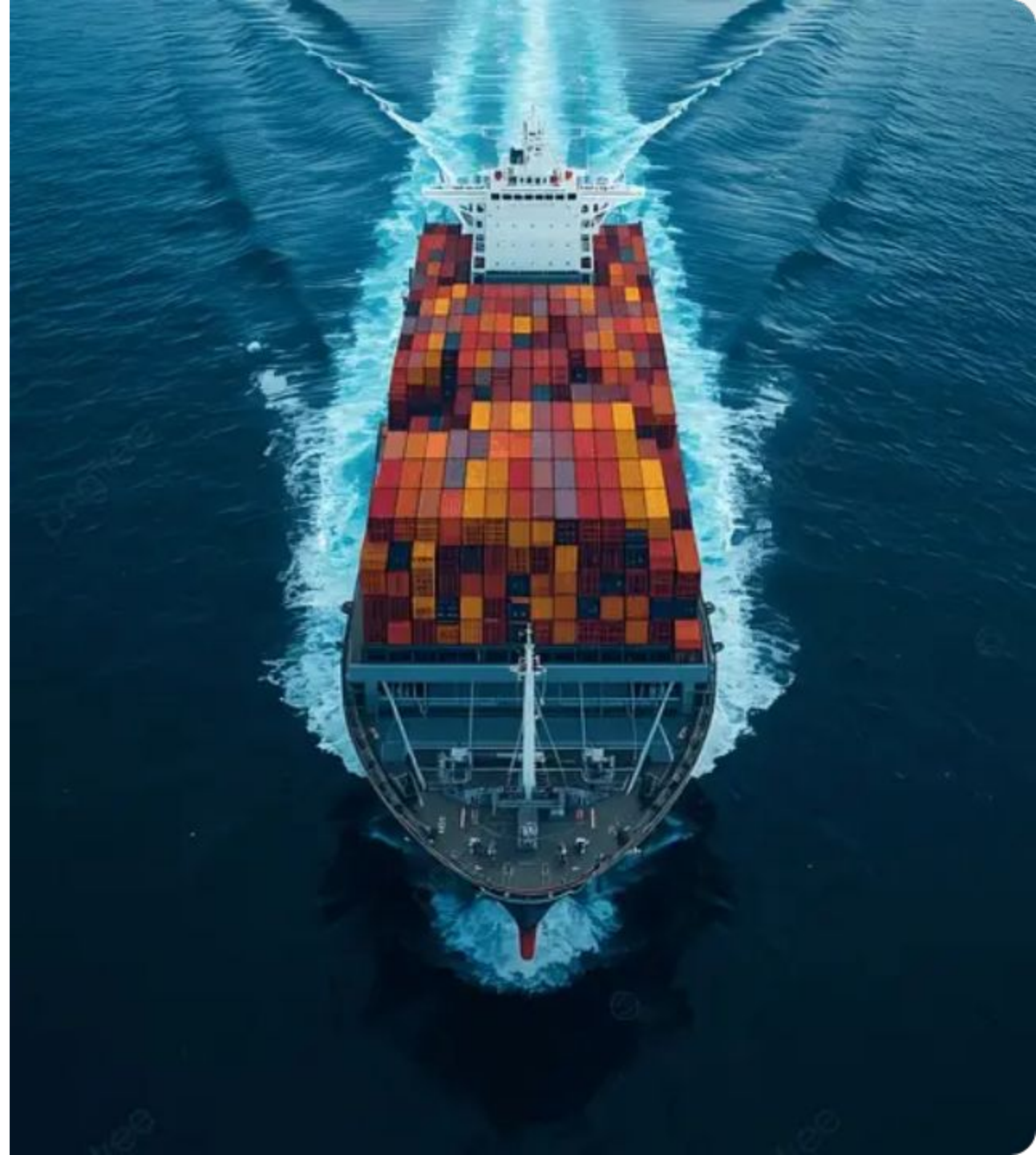
REGPULSE AI AUTOMATED ECOSYSTEM

Translating complex news on regulatory matters into clear, streamlined executive audio briefings.

RegPulse AI acts as a smart autonomous pipeline that converts cold, highly technical and legal news into live, professional business audio summaries.

Combining localized web-scraping with structured OpenAI intelligence and high-fidelity narration models to keep crews and executives perfectly informed.

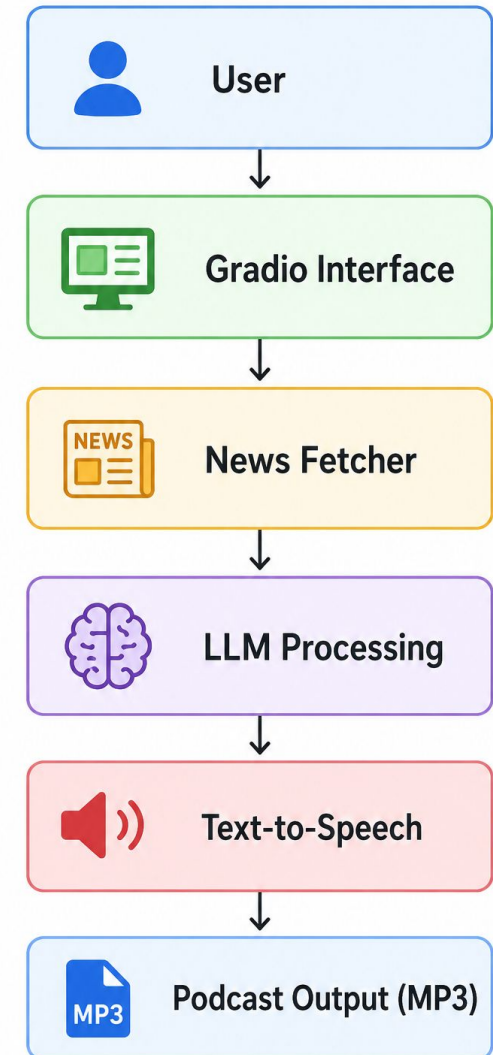
EU Maritime Intelligence System • Model MVP



MODULAR SYSTEM DESIGN

A decoupled system architecture allows for high scalability and rapid debug workflows. This structured data flow transforms raw documents into podcast files:

Component	Module Responsibility	Programming Interface (API)	Core Data Output
News Fetcher	Extracts country-specific policies (Andreas)	NEWS API	news_text (Structured Text)
LLM Processor	Prompt engineering & script structuring (Gretel)	OpenAI gpt-4o-mini	Structured JSON Podcast Script
TTS Generator	Multi-voice high-fidelity vocal synthesis (Vincent)	OpenAI tts-1 (alloy & echo)	podcast.mp3 (Merged Audio)



TECHNICAL CHOICES

Why these technologies?

Python

- Flexible for AI integration
- Large ecosystem
- Easy API communication

OpenAI API

- Strong language understanding
- Reliable summarization
- Natural text generation

OpenAI Text-to-Speech

- Converts scripts into realistic audio
- Easy API integration

Gradio

- Fast creation of AI interfaces
- Easy demonstration
- User-friendly

CHALLENGES ENCOUNTERED

Integrating independent modules

- We defined clear interfaces between modules and tested each component independently before integrating the complete workflow. This made debugging easier and reduced integration issues.

Retrieving relevant maritime news

- We realized that improving the Boolean query was not enough. We redesigned the retrieval strategy by combining broader searches with local filtering, relevance scoring and article ranking before sending the results to the LLM.

Prompt Engineering

- We iteratively refined our prompts to produce structured, consistent outputs focused on executive summaries, business impact and podcast-ready scripts.

Building an end-to-end workflow

- We connected the News Retrieval, LLM Processing, Text-to-Speech and Gradio modules into a single pipeline with consistent data flow and error handling.

Improving the user experience

- We enhanced the Gradio interface using layouts, HTML and CSS to create a cleaner executive dashboard without adding unnecessary frontend complexity.

Working under tight time constraints

- We continuously prioritized high-impact improvements, focused on delivering a stable MVP first, and left more advanced features for future iterations.

Takeaway: Resolving these integration errors in real time provided invaluable, hands-on experience in collaborative, software development

SYSTEM WORKFLOW

1

User Input

The user selects: An EU country, A timeframe, Last Week, Last Month

2

News Retrieval

The system searches for recent maritime environmental and regulatory news related to the selected country. Relevant news articles are collected and prepared for processing.

3

AI Processing congue tempus

OpenAI analyzes the retrieved articles and generates:

1. Executive Summary
2. Business Impact
3. Podcast Script

Rather than simply shortening the articles, the AI restructures the information into a professional executive briefing suitable for business users.

SYSTEM WORKFLOW

4

Text-to-Speech

The generated podcast script is sent to the OpenAI Text-to-Speech API. The application automatically produces an MP3 audio file containing the podcast narration.

5

User Interface

The Gradio web interface displays:

1. Executive Summary
2. Business Impact
3. Podcast Script
4. Downloadable and playable MP3

6

Final Output

The user receives both a written executive briefing and an AI-generated podcast summarizing the latest maritime environmental developments.



RegPulse AI
EXECUTIVE DAILY PODCAST & INTELLIGENCE

LIVE DEMO

FUTURE IMPROVEMENTS

Possible Future Enhancements

- Expand support to additional countries and regions
- Generate podcasts in multiple languages
- Integrate higher-quality AI voices for a more natural listening experience
- Schedule automatic daily podcast generation
- Deploy the application to the cloud for broader accessibility
- Add user authentication and personalized profiles
- Create a searchable archive of historical podcasts
- Integrate additional news sources to improve coverage and reliability
- Enhance timeframe selection with more flexible date ranges
- Add advanced filtering and search capabilities
- Send daily digest emails or push notifications with the latest podcast updates
- Accommodate integration with paid news platforms.

TAKEAWAYS

Key Learnings

- AI systems are built by combining multiple specialised components
- Good project structure makes integration easier
- Prompt design strongly influences AI output quality
- Testing each module separately reduces integration problems
- AI can transform complex information into accessible formats

Questions & Answers

Thank you for your attention. RegPulse AI is ready to revolutionize maritime environmental compliance.

Project Repository github.com/ironhack-regpulse-ai

IMAGE SOURCES



https://png.pngtree.com/thumb_back/fw800/background/20251205/pngtree-aerial-view-of-a-large-container-ship-navigating-the-deep-blue-image_20656381.w

ebp

Source: [pngtree.com](https://png.pngtree.com)